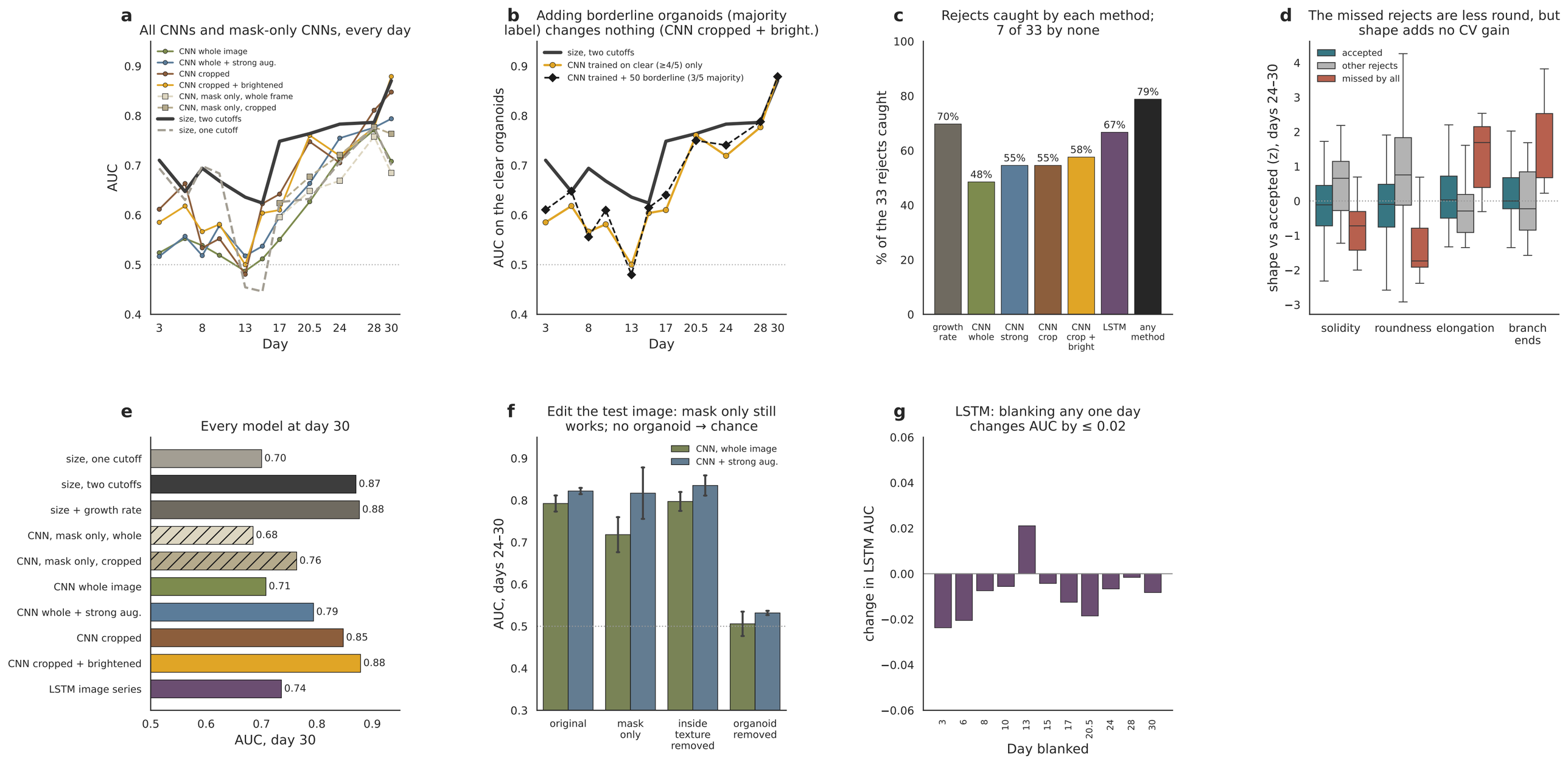
a
All CNNs and mask-only CNNs, every day
CNN whole image
CNN whole + strong aug.
CNN cropped
CNN cropped + brightened
CNN, mask only, whole frame
CNN, mask only, cropped
size, two cutoffs
size, one cutoff
AUC
Day
b
Adding borderline organoids (majority label) changes nothing (CNN cropped + bright.)
size, two cutoffs
CNN trained on clear (≥4/5) only
CNN trained + 50 borderline (3/5 majority)
AUC on the clear organoids
Day
c
Rejects caught by each method; 7 of 33 by none
% of the 33 rejects caught
70%
48%
55%
55%
58%
67%
79%
growth rate
CNN whole
CNN strong
CNN crop
CNN crop + bright
LSTM
any method
d
The missed rejects are less round, but shape adds no CV gain
accepted
other rejects
missed by all
shape vs accepted (z), days 24–30
solidity
roundness
elongation
branch ends
e
Every model at day 30
size, one cutoff
size, two cutoffs
size + growth rate
CNN, mask only, whole
CNN, mask only, cropped
CNN whole image
CNN whole + strong aug.
CNN cropped
CNN cropped + brightened
LSTM image series
0.70
0.87
0.88
0.68
0.76
0.71
0.79
0.85
0.88
0.74
AUC, day 30
f
Edit the test image: mask only still works; no organoid → chance
CNN, whole image
CNN + strong aug.
AUC, days 24–30
original
mask only
inside texture removed
organoid removed
g
LSTM: blanking any one day changes AUC by ≤ 0.02
change in LSTM AUC
Day blanked